Building and evaluation of a PBPK model for Clomiphene in adults

Version	**vevaluation**
based on *Model Snapshot* and *Evaluation Plan*	https://github.com/Open-Systems-Pharmacology/Clomiphene-Model/releases/tag/vevaluation
OSP Version	evaluation
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This evaluation report and the corresponding PK-Sim project file are filed at:

https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/

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1 Introduction

(E)-Clomiphene is the trans-isomer of clomiphene and contributes to the clinical pharmacology of clomiphene citrate, a selective estrogen receptor modulator used in reproductive medicine. Its pharmacokinetics are complex because clomiphene is administered as an E/Z mixture, undergoes sequential oxidative metabolism and shows clinically relevant variability linked to CYP2D6 activity [1].

The model describes (E)-clomiphene as a whole-body parent-metabolite PBPK model with (E)-N-desethylclomiphene, (E)-4-hydroxyclomiphene and (E)-4-hydroxy-N-desethylclomiphene as measured metabolites. This structure supports evaluation of parent and metabolite plasma concentration-time data after oral clomiphene or enclomiphene administration and enables activity-score-dependent CYP2D6 DGI simulations.

The clinical evaluation covers adult plasma pharmacokinetic data after single-dose and multiple-dose oral administration. The dataset includes a CYP2D6 activity-score stratified panel study with parent and metabolite measurements and external literature studies reporting (E)-clomiphene concentration-time data. The model was developed for clomiphene DGI and DDGI scenario prediction and was subsequently integrated into the broader CYP2D6 interaction network [1, 2].

2 Methods

2.1 Modeling strategy

The PBPK model was developed using the Open Systems Pharmacology Suite. Distribution and elimination processes were implemented in a whole-body model structure using standard physiological organ compartments and compound-specific physicochemical and biochemical input parameters.

Model development followed an iterative workflow in which in vitro and physicochemical information was combined with clinical plasma concentration-time data. Parameters with direct literature support were used as fixed inputs where possible, while selected parameters controlling oral absorption, metabolism, residual clearance and activity-score-dependent CYP2D6 scaling were optimized against clinical data.

The model represents (E)-clomiphene and three metabolites as separate compounds. (E)-N-Desethylclomiphene is included as the main desethyl metabolite, while (E)-4-hydroxyclomiphene and (E)-4-hydroxy-N-desethylclomiphene are included as active hydroxylated metabolites. The four-compound structure is required because the available panel study reports parent and metabolite concentration-time data across CYP2D6 activity-score groups.

The clinical panel study by Kovar et al. informed activity-score-dependent parent-metabolite disposition after oral clomiphene citrate administration [1]. Literature studies by Mikkelson et al., Ratiopharm GmbH, Wiehle et al. and Miller et al. provided additional single-dose and multiple-dose oral concentration-time data for (E)-clomiphene [3-6]. The literature studies are used to evaluate oral formulation behavior and the parent compound over a dose range broader than the DGI panel study.

The major proteins represented in the model are CYP2D6, CYP3A4 and CYP2B6. CYP2D6 accounts for the activity-score-dependent formation and elimination pathways that drive DGI behavior. CYP3A4 contributes to desethylation and metabolite turnover, while CYP2B6 contributes to (E)-4-hydroxyclomiphene formation. Renal filtration, enterohepatic recirculation and residual hepatic clearance are retained where required to describe parent and metabolite disposition.

2.2 Data used

2.2.1 In vitro and physicochemical data

Drug-dependent input parameters for (E)-clomiphene and its metabolites were taken from the published clomiphene PBPK model and the cited primary sources therein [1]. The key physicochemical properties include molecular weight, pK_a, lipophilicity, solubility, fraction unbound in plasma and intestinal permeability. Compound-specific values are used for the parent and each metabolite because polarity, protein binding and elimination differ across the metabolic sequence.

CYP2D6 K_m values are based on in vitro metabolism data for clomiphene and its metabolites, with nonspecific in vitro binding accounted for in the model development workflow [1, 7]. Activity-score-dependent k_{cat} values were identified for normal metabolizers and scaled to additional CYP2D6 activity-score groups using pathway-specific in vitro scaling factors [1, 2].

Parameter group	Model use	Main source
Molecular weight, pK_a, solubility, lipophilicity and f_u	Parent and metabolite physicochemical and distribution inputs	Kovar et al. 2022 and cited primary sources
CYP2D6 K_m and k_{cat} values	activity-score-dependent hydroxylation and downstream metabolite clearance	Mürdter et al. 2012, Kovar et al. 2022
CYP3A4 and CYP2B6 metabolic pathways	Desethylation, hydroxylation and metabolite turnover	Mazzarino et al. 2013, Kovar et al. 2022
Plasma binding estimates	Distribution and unbound concentration scaling	Watanabe et al. 2018, Kovar et al. 2022
Partition coefficients and permeability	Tissue distribution and cellular permeability	Schmitt 2008, Rodgers and Rowland 2005, Rodgers et al. 2006, Berezhkovskiy 2004

2.2.2 Clinical data

Clinical plasma concentration-time data were used for model building and evaluation after oral clomiphene or enclomiphene administration. The clinical dataset includes parent (E)-clomiphene observations from external literature and parent-metabolite observations from the CYP2D6 activity-score panel study.

Publication	Clinical data used
Kovar et al. 2022	Oral single-dose clomiphene citrate data in healthy adult premenopausal women with CYP2D6 activity-score groups were used for parent-metabolite DGI evaluation.
Mikkelson et al. 1986	Oral single-dose clomiphene citrate data in healthy adult volunteers supported evaluation of (E)-clomiphene disposition after a low single dose.
Ratiopharm GmbH 2016	Oral single-dose clomiphene citrate data from the product information dataset supported an additional (E)-clomiphene single-dose comparison.
Wiehle et al. 2013	Oral multiple-dose enclomiphene citrate data in adult men with secondary hypogonadism supported evaluation across 6.25, 12.5 and 25 mg dose levels.
Miller et al. 2019	Oral multiple-dose clomiphene data in adult men supported an additional repeated-dose (E)-clomiphene comparison.

2.3 Model parameters and assumptions

2.3.1 Absorption

The model includes oral administration of (E)-clomiphene after clomiphene citrate or enclomiphene citrate dosing. The clinical panel study used clomiphene citrate tablets containing the E/Z isomer mixture, and the model evaluates the (E)-clomiphene fraction relevant to the measured parent-metabolite dataset [1].

Oral absorption is represented with a tablet Weibull dissolution function and compound intestinal permeability. The tablet formulation parameters describe the oral input profile, while differences between CYP2D6 activity-score groups are assigned to metabolism rather than absorption.

The external literature studies include single-dose and multiple-dose oral dosing conditions. These data support evaluation of oral input behavior across lower single doses and repeated enclomiphene exposure, but they do not by themselves resolve individual CYP2D6 activity in the absence of genotype or phenotype information.

2.3.2 Distribution

(E)-Clomiphene is represented as a highly lipophilic, highly protein-bound compound with logP of 5.67 and f_u of 0.08% as summarized in Section 2.2. The metabolites are less lipophilic but remain substantially bound, with logP values of 4.17 for (E)-N-desethylclomiphene, 5.50 for (E)-4-hydroxyclomiphene and 3.71 for (E)-4-hydroxy-N-desethylclomiphene [1].

Partition coefficients were calculated with Schmitt, Rodgers and Rowland or Berezhkovskiy methods depending on the compound. The model therefore treats the parent and metabolites as separate distribution entities rather than as a single lumped active moiety.

The active hydroxylated metabolites are represented with their own molecular weight, pK_a, lipophilicity, plasma binding and clearance parameters. This separation is required because metabolite exposure is controlled by both formation from upstream compounds and metabolite-specific elimination.

2.3.3 Metabolism and elimination

Clomiphene elimination is represented by CYP2D6-dependent hydroxylation and desethylation, CYP3A4-dependent desethylation and metabolite turnover, CYP2B6-dependent hydroxylation, renal filtration, enterohepatic recirculation and residual hepatic clearance components.

- CYP2D6

CYP2D6 is the central enzyme for DGI behavior in the model. It forms (E)-4-hydroxyclomiphene from (E)-clomiphene and contributes to desethylation and downstream metabolite elimination. Pathway-specific K_m values are used, while k_{cat} values depend on activity score [1, 7].

The CYP2D6 activity-score implementation is the key determinant of simulated exposure differences across poor, intermediate, normal and ultrarapid metabolizer groups. Poor-metabolizer activity is set to zero, while non-zero activity-score groups use scaled k_{cat} values.

- CYP3A4, CYP2B6 and residual clearance

CYP3A4 is implemented for (E)-N-desethylclomiphene formation and downstream metabolite turnover. CYP2B6 contributes to (E)-4-hydroxyclomiphene formation from (E)-clomiphene. These pathways support non-CYP2D6 clearance and metabolite formation where the in vitro data indicate additional enzymatic contribution [1, 9].

Residual hepatic clearance terms are empirical. They should be interpreted as structural model components required to describe total disposition rather than as direct measurements of a single biochemical pathway.

- Renal filtration and enterohepatic recirculation

Renal filtration is included with compound-specific GFR fractions for the parent and metabolites. Enterohepatic recirculation is represented with continuous bile release fractions where required by the model structure.

2.3.4 Automated parameter identification

The following parameters were optimized by fitting the model to clinical data:

Model parameter
CYP2D6 k_{cat} values for parent and metabolite pathways
CYP3A4 and CYP2B6 k_{cat} values
Residual hepatic clearance terms
GFR fractions
Lipophilicity and f_u values where direct source support was insufficient
Specific intestinal permeability and tablet formulation parameters

The optimized parameters were selected to describe oral formulation behavior, CYP2D6 activity-score-dependent parent-metabolite disposition and residual metabolite elimination across the available clinical data. Literature-supported physicochemical inputs and binding parameters were retained as fixed model inputs where possible.

3 Results and Discussion

The evaluation combines goodness-of-fit diagnostics and concentration-time profiles for (E)-clomiphene, (E)-4-hydroxyclomiphene, (E)-N-desethylclomiphene and (E)-4-hydroxy-N-desethylclomiphene. The diagnostic plots are separated by analyte so that parent and metabolite observations can be assessed without ambiguous legends.

Overall model performance should be interpreted across the full clinical dataset rather than from a single study arm. The external literature studies primarily assess oral parent-compound disposition across single-dose and multiple-dose conditions, while the CYP2D6 activity-score panel study assesses parent-metabolite behavior across CYP2D6 activity groups.

The clomiphene model publication reported that the model described the observed parent and metabolite concentration-time profiles and supported prediction of CYP2D6 DGI and CYP2D6/CYP3A4 DDGI scenarios [1]. The present report reproduces the concentration-time and GOF assessment structure for the model repository and keeps parent and metabolite endpoints separated in the diagnostic sections.

Residual uncertainty remains for the exact contribution of non-CYP2D6 pathways, protein-binding estimates and individual variability within the same CYP2D6 activity-score group. The model therefore supports CYP2D6 activity-score-based exposure evaluation, while further genotype-specific refinement would require additional clinical data with paired parent-metabolite profiles.

3.1 Clomiphene final input parameters

The following input-parameter tables summarize the final model parameterization for (E)-clomiphene and its metabolites. Parameter interpretation should be based on the source descriptions in Section 2.2 and the assumptions in Section 2.3.

Compound: (E)-clomiphene

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	13.8 mg/l	Publication-Das et al. 2020	Das et al. 2020 pH 6.8	True
Reference pH	6.8	Publication-Das et al. 2020	Das et al. 2020 pH 6.8	True
Lipophilicity	5.67 Log Units	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Fraction unbound (plasma, reference value)	0.00084694454938	Parameter Identification-Parameter Identification-Optimized	Measurement	True
Specific intestinal permeability (transcellular)	0.0828652784 cm/min	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Cl	1	Publication-Siramshetty et al. 2022		
Is small molecule	Yes			
Molecular weight	405.96 g/mol	Publication-Siramshetty et al. 2022		
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Schmitt
Cellular permeabilities	Charge dependent Schmitt

Processes

Metabolizing Enzyme: CYP3A4-Mürdter et al. 2012

Molecule: CYP3A4

Metabolite: (E)-Deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	44.9466036298 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=2) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	306.3819989852 1/min	Parameter Identification-Parameter Identification-Optimized

Systemic Process: Glomerular Filtration-Pharmacokinetic panel study

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	0.9183035142	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2B6-Mürdter et al. 2012

Molecule: CYP2B6

Metabolite: 4-Hydroxy-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	39 pmol/mg mic. protein	Unknown
Km	0.600817495 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	7.493654322 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=2) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	130.3516139745 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (AS=3) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	171.5386005 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=0.5) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	22.68117655 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=0.75) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	29.84299905 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=0) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	0 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=1) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	66.12599871 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=3) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	184.2542995 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (AS=2) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	121.3558299218 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (AS=0) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	0 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (AS=0.5) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	21.11591042 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (AS=0.75) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	27.7834835 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (AS=1) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	61.56253235 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	0 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0.5) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	57.48197455 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0.75) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	81.56688504 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=1) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	175.0793273 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=3) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	467.2288197 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (Miller 2018) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	7.1740592157 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (Miller 2018) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	7.7058530943 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (Miller 2018) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	18.1120478905 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (Wiehle 2013 12.5 mg) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	52.779552089 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (Wiehle 2013 12.5 mg) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	49.1371464562 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (Wiehle 2013 12.5 mg) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	124.0545029059 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (Wiehle 2013 25 mg) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	18.4144257821 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (Wiehle 2013 6.25 mg) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	37.3233521148 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (Study Ratiopharm) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	120.4548809775 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (Mikkelson 1986) - (Z)-3-OH-clomiphene

Molecule: CYP2D6

Metabolite: (Z)-3-OH-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	38.2 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.030872045 µmol/l	Publication-In Vitro-Kröner 2018
kcat	90.6375098012 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (Wiehle 2013 25 mg) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	17.143615297 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (Wiehle 2013 6.25 mg) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	34.7476048301 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (Study Ratiopharm) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	112.1420871092 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mazzarino et al. 2013 (Mikkelson 1986) - (E)-DE-clomiphene

Molecule: CYP2D6

Metabolite: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	290.7 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.783674994 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	84.3824628525 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (Wiehle 2013 25 mg) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	43.2817700468 1/min	Unknown

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (Wiehle 2013 6.25 mg) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	87.725827714 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (Study Ratiopharm) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	283.1204470445 1/min	Unknown

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (Mikkelson 1986) - (E)-4-OH-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	90.1 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.130612499 µmol/l	Publication-In Vitro-Kröner 2018
kcat	213.0368822389 1/min	Unknown

Compound: (E)-N-desethylclomiphene

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	0.46 mg/ml	Database-Chemicalize	Measurement	True
Reference pH	6.5	Database-Chemicalize	Measurement	True
Lipophilicity	4.1726909096 Log Units	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Fraction unbound (plasma, reference value)	0.0085575590017	Parameter Identification-Parameter Identification-Optimized	Measurement	True
Cl	1			
Is small molecule	Yes			
Molecular weight	377.91 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	Charge dependent Schmitt

Processes

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=3) - (E)-4-Hydroxy-deethyl-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-Hydroxy-deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	53.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.48716284 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	84.00858577 1/min	Parameter Identification-Parameter Identification-Optimized

Systemic Process: Glomerular Filtration-Pharmacokinetic panel study

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	0.1005414757	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=3) - (E)-N,N-didesethylclomiphene

Molecule: CYP2D6

Metabolite: (E)-N,N-didesethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	20.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.96838467 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	8.248111562 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP3A4-Mürdter et al. 2012 - (E)-N,N-didesethylclomiphene

Molecule: CYP3A4

Metabolite: (E)-N,N-didesethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	20.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	0.96838467 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	0.7771909271 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0) - (E)-N,N-didesethylclomiphene

Molecule: CYP2D6

Metabolite: (E)-N,N-didesethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	20.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.96838467 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	0 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0.5) - (E)-N,N-didesethylclomiphene

Molecule: CYP2D6

Metabolite: (E)-N,N-didesethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	20.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.96838467 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	1.01531891 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0.75) - (E)-N,N-didesethylclomiphene

Molecule: CYP2D6

Metabolite: (E)-N,N-didesethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	20.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.96838467 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	1.335916645 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=1) - (E)-N,N-didesethylclomiphene

Molecule: CYP2D6

Metabolite: (E)-N,N-didesethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	20.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.96838467 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	2.960118793 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=2) - (E)-N,N-didesethylclomiphene

Molecule: CYP2D6

Metabolite: (E)-N,N-didesethylclomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	20.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.96838467 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	5.8351672527 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=2) - (E)-4-Hydroxy-deethyl-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-Hydroxy-deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	53.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.48716284 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	64.5203767233 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0.5) - (E)-4-Hydroxy-deethyl-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-Hydroxy-deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	53.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.48716284 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	10.34807199 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0.75) - (E)-4-Hydroxy-deethyl-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-Hydroxy-deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	53.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.48716284 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	12.36587928 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=1) - (E)-4-Hydroxy-deethyl-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-Hydroxy-deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	53.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.48716284 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	28.59140514 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Mürdter et al. 2012 (AS=0) - (E)-4-Hydroxy-deethyl-clomiphene

Molecule: CYP2D6

Metabolite: (E)-4-Hydroxy-deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	53.6 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.48716284 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	0 1/min	Parameter Identification-Parameter Identification-Optimized

Compound: (E)-4-hydroxyclomiphene

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	0.06 mg/ml		Measurement	True
Reference pH	6.5		Measurement	True
Lipophilicity	5.5012508036 Log Units	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Fraction unbound (plasma, reference value)	0.00449	Parameter Identification-Parameter Identification-Optimized	Measurement	True
Cl	1			
Is small molecule	Yes			
Molecular weight	421.97 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Berezhkovskiy
Cellular permeabilities	PK-Sim Standard

Processes

Metabolizing Enzyme: CYP3A4-Mürdter et al. 2012

Molecule: CYP3A4

Metabolite: (E)-4-Hydroxy-deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	161.2 pmol/min/mg mic. protein	Publication-In Vitro-Mürdter et al. 2012
Km	3.400160633 µmol/l	Publication-In Vitro-Mürdter et al. 2012
kcat	19.52 1/min	Parameter Identification-Parameter Identification-Optimized

Systemic Process: Glomerular Filtration-Pharmacokinetic panel study

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	0.2386673303	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=3)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	3.59784439 µmol/l	Publication-In Vitro-Kröner 2018
kcat	1208.794021 1/min	Parameter Identification-Parameter Identification-Optimized

Systemic Process: Total Hepatic Clearance-Assumption

Species: Human

Parameters

Name	Value	Value Origin
Fraction unbound (experiment)	0.0133	
Lipophilicity (experiment)	4.6502520184 Log Units	
Plasma clearance	0 ml/min/kg	
Specific clearance	23.7771867087 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=0)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	3.59784439 µmol/l	Publication-In Vitro-Kröner 2018
kcat	0 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=0.5)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	3.59784439 µmol/l	Publication-In Vitro-Kröner 2018
kcat	148.7990819 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=0.75)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	3.59784439 µmol/l	Publication-In Vitro-Kröner 2018
kcat	195.7839731 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=1)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	3.59784439 µmol/l	Publication-In Vitro-Kröner 2018
kcat	433.8173497 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Kröner 2018 (AS=2)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	3.59784439 µmol/l	Publication-In Vitro-Kröner 2018
kcat	855.1672990338 1/min	Parameter Identification-Parameter Identification-Optimized

Compound: (E)-4-hydroxy-N-desethylclomiphene

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	0.17 mg/ml	Database-Chemicalize	Measurement	True
Reference pH	6.5	Database-Chemicalize	Measurement	True
Lipophilicity	3.7147097639 Log Units	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Fraction unbound (plasma, reference value)	0.0132	Publication-Other-Watanabe et al. 2018	Measurement	True
Cl	1			
Is small molecule	Yes			
Molecular weight	393.91 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Schmitt
Cellular permeabilities	Charge dependent Schmitt

Processes

Systemic Process: Total Hepatic Clearance-Assumption

Species: Human

Parameters

Name	Value	Value Origin
Fraction unbound (experiment)	0.0132	
Lipophilicity (experiment)	6 Log Units	
Plasma clearance	0 ml/min/kg	
Specific clearance	8.5014115561 1/min	Parameter Identification-Parameter Identification-Optimized

Systemic Process: Glomerular Filtration-Pharmacokinetic panel study

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	0.1294297124	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Assumption (AS=3)

Molecule: CYP2D6

Metabolite: (Z)-3,4-Dihydroxy-deethyl-clomiphene

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.855527731 µmol/l	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
kcat	299.2603852 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Assumption (AS=0)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.855527731 µmol/l	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
kcat	0 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Assumption (AS=0.5)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.855527731 µmol/l	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
kcat	36.83809632 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Assumption (AS=0.75)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.855527731 µmol/l	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
kcat	48.47011666 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Assumption (AS=1)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.855527731 µmol/l	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
kcat	107.3998919 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Assumption (AS=2)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	474.5 pmol/min/mg mic. protein	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.855527731 µmol/l	Publication-In Vitro-Kröner 2018 - (E)-4-OH-clomiphene
kcat	211.7132372293 1/min	Parameter Identification-Parameter Identification-Optimized

3.2 Diagnostic plots

Goodness-of-fit diagnostics are shown separately for each analyte. This layout keeps parent and metabolite observations visually distinct and supports interpretation of CYP2D6 activity-score-dependent parent-metabolite behavior.

3.2.1 (E)-Clomiphene goodness-of-fit diagnostics

(E)-Clomiphene GOF diagnostics include the external literature studies and the CYP2D6 activity-score panel study. These diagnostics assess parent-compound exposure across oral single-dose, oral multiple-dose, and CYP2D6-stratified conditions.

Table 3-1: GMFE for (E)-Clomiphene observed versus simulated plasma concentration-time data

Group	GMFE
(E)-clomiphene	1.44

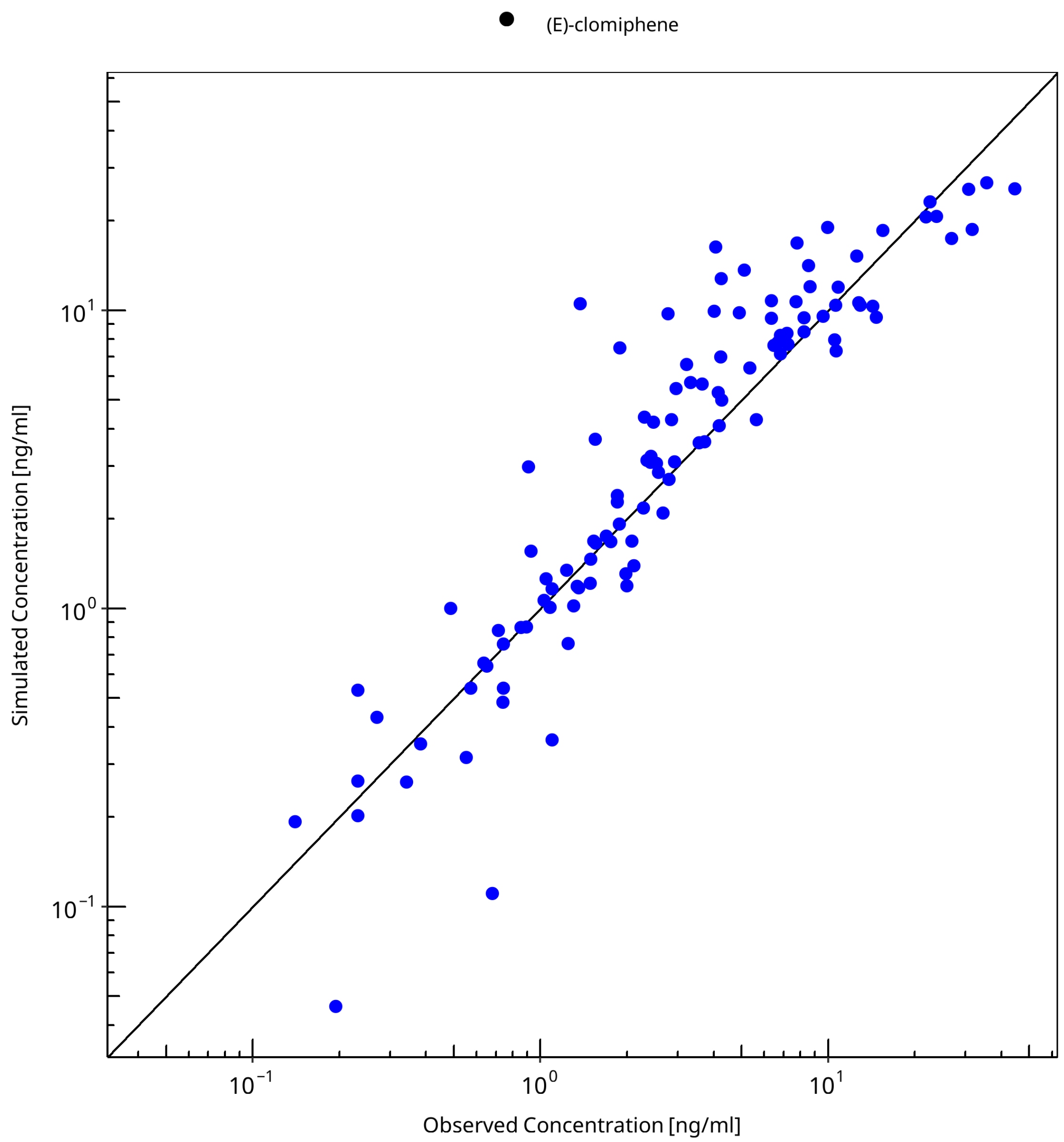

Figure 3-1: (E)-Clomiphene observed versus simulated plasma concentration-time data

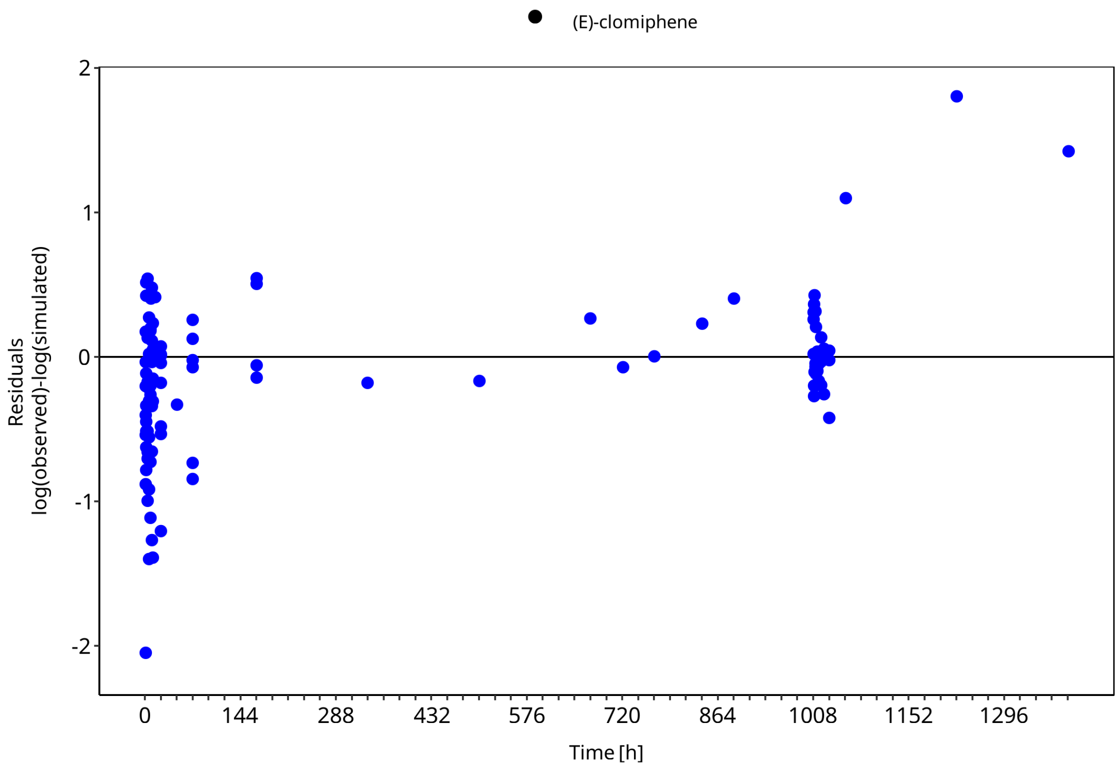

Figure 3-2: (E)-Clomiphene observed versus simulated plasma concentration-time data

3.2.2 (E)-4-Hydroxyclomiphene goodness-of-fit diagnostics

(E)-4-Hydroxyclomiphene GOF diagnostics focus on the CYP2D6 activity-score panel study. This metabolite is directly informative for CYP2D6-dependent hydroxylation of (E)-clomiphene.

Table 3-2: GMFE for (E)-4-Hydroxyclomiphene observed versus simulated plasma concentration-time data

Group	GMFE
(E)-4-hydroxyclomiphene	1.64

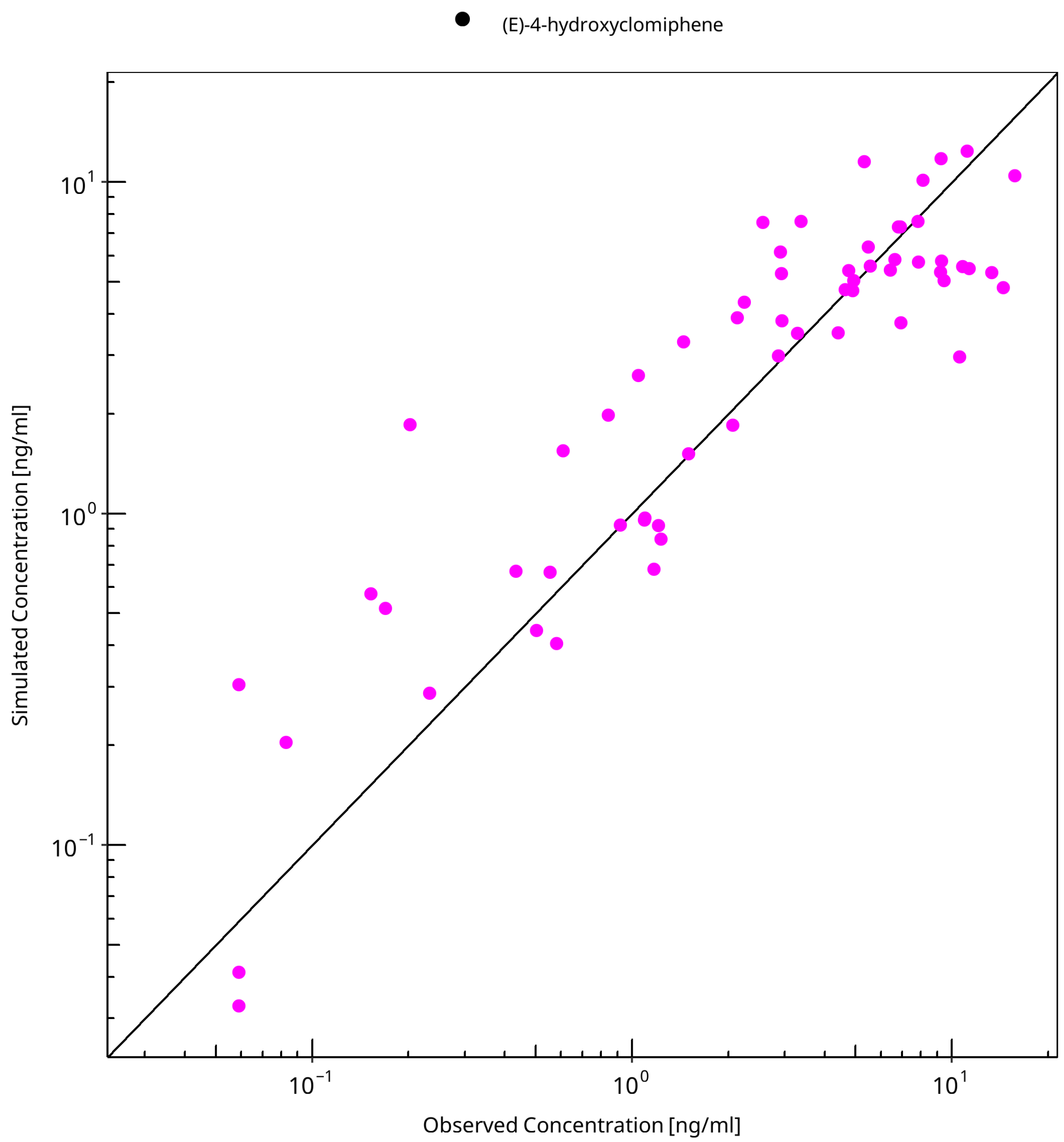

Figure 3-3: (E)-4-Hydroxyclomiphene observed versus simulated plasma concentration-time data

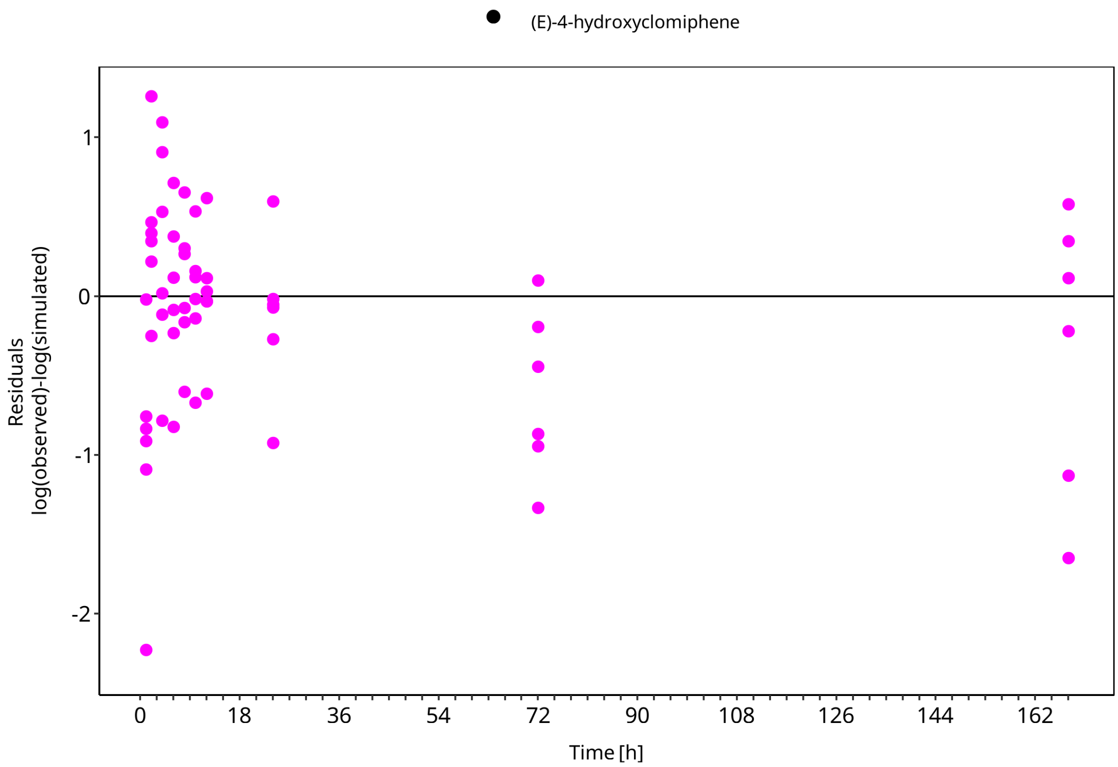

Figure 3-4: (E)-4-Hydroxyclomiphene observed versus simulated plasma concentration-time data

3.2.3 (E)-N-Desethylclomiphene goodness-of-fit diagnostics

(E)-N-Desethylclomiphene GOF diagnostics focus on the CYP2D6 activity-score panel study. This metabolite reflects desethylation contributions from CYP2D6 and CYP3A4 pathways.

Table 3-3: GMFE for (E)-N-Desethylclomiphene observed versus simulated plasma concentration-time data

Group	GMFE
(E)-N-desethylclomiphene	1.87

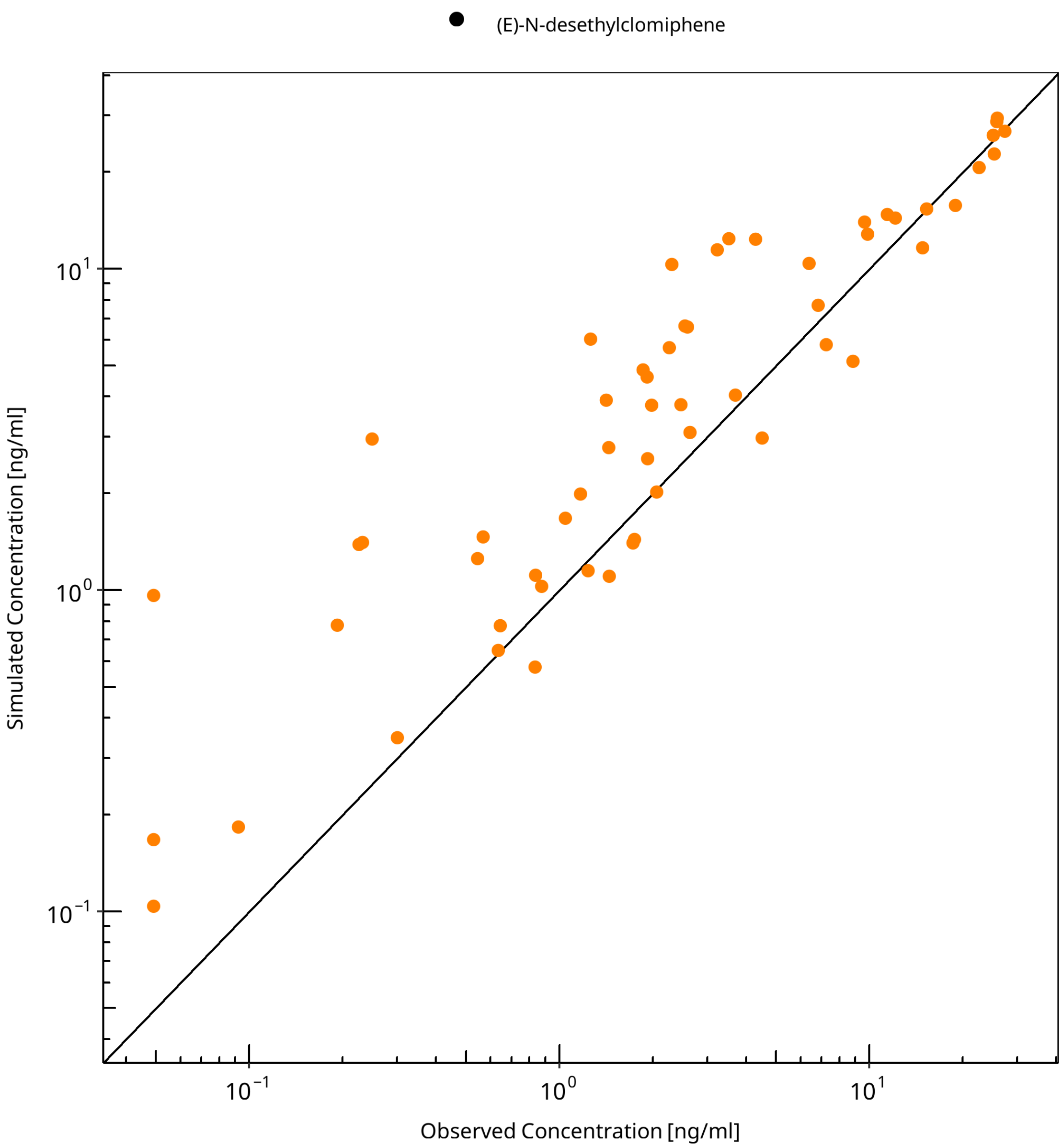

Figure 3-5: (E)-N-Desethylclomiphene observed versus simulated plasma concentration-time data

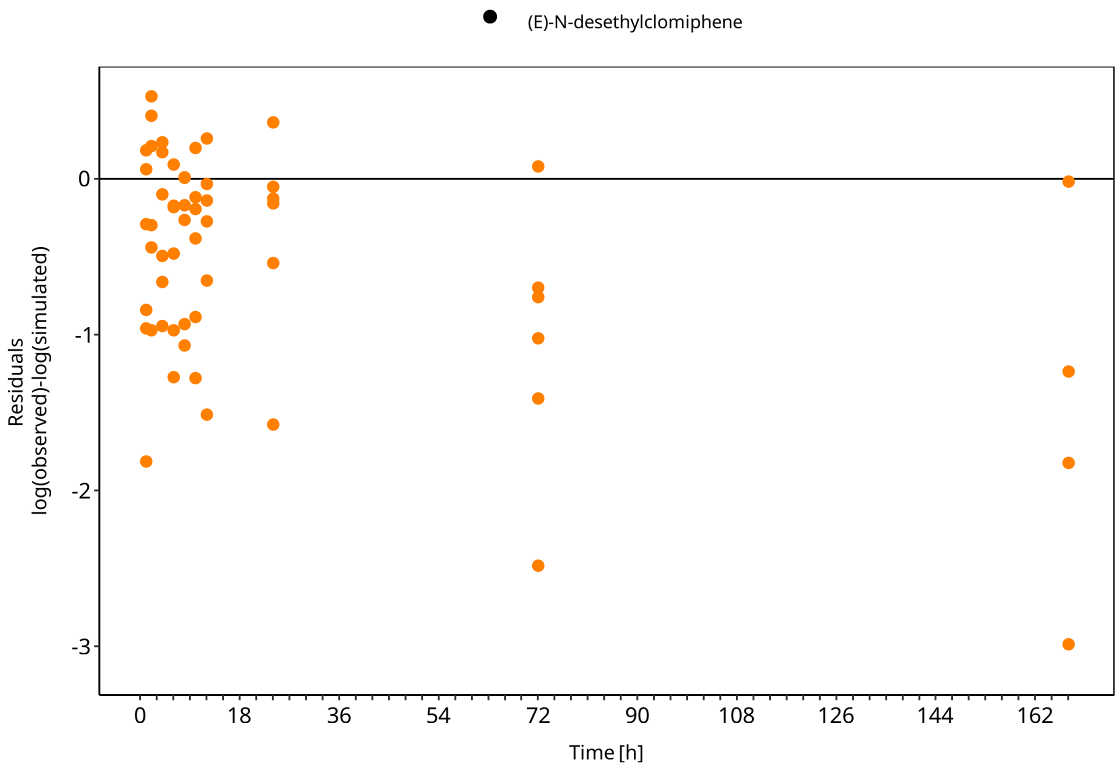

Figure 3-6: (E)-N-Desethylclomiphene observed versus simulated plasma concentration-time data

3.2.4 (E)-4-Hydroxy-N-desethylclomiphene goodness-of-fit diagnostics

(E)-4-Hydroxy-N-desethylclomiphene GOF diagnostics focus on the CYP2D6 activity-score panel study. This section supports evaluation of downstream metabolite formation and elimination.

Table 3-4: GMFE for (E)-4-Hydroxy-N-desethylclomiphene observed versus simulated plasma concentration-time data

Group	GMFE
(E)-4-hydroxy-N-desethylclomiphene	1.56

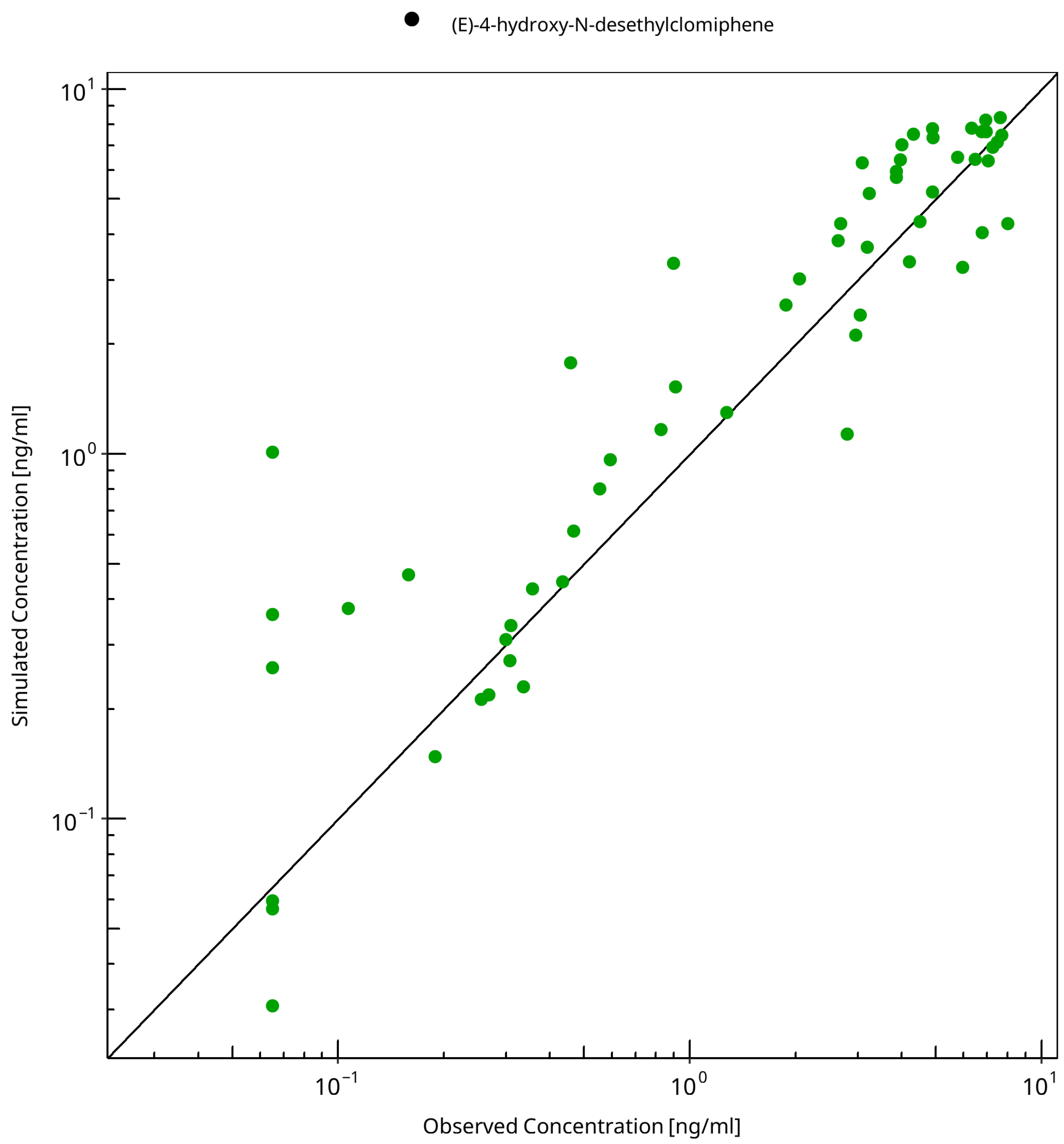

Figure 3-7: (E)-4-Hydroxy-N-desethylclomiphene observed versus simulated plasma concentration-time data

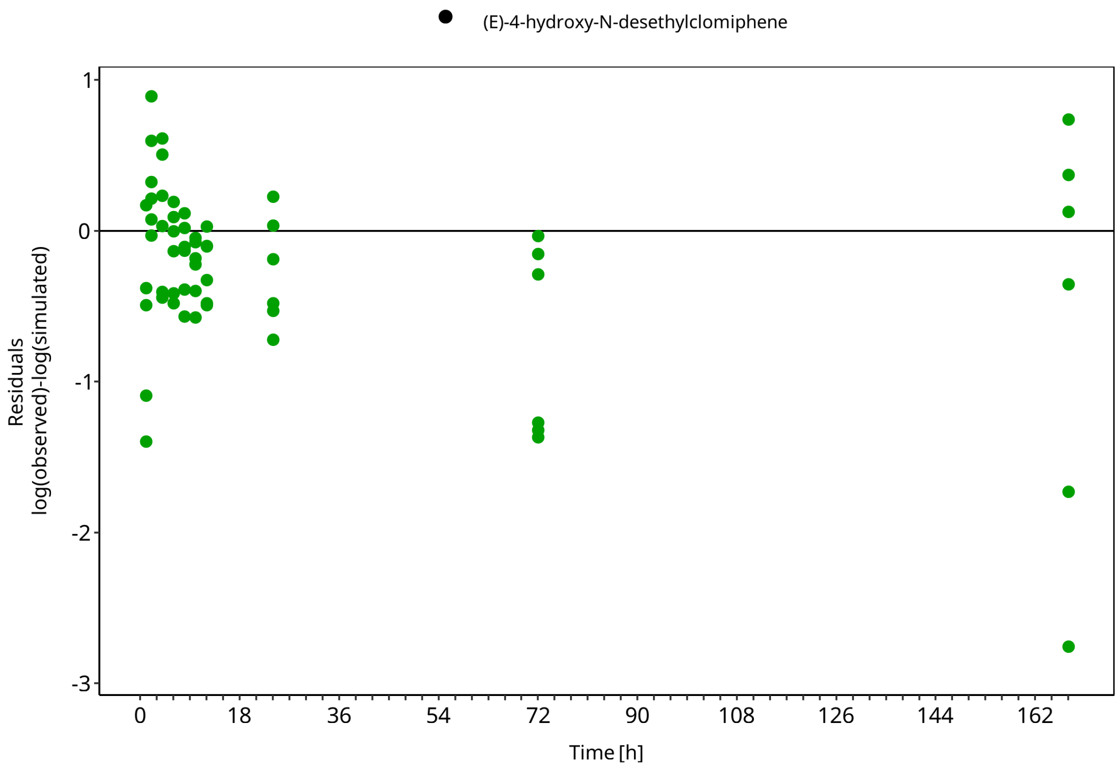

Figure 3-8: (E)-4-Hydroxy-N-desethylclomiphene observed versus simulated plasma concentration-time data

3.3 Concentration-time profiles

The concentration-time profile sections show observed and simulated plasma concentrations for the CYP2D6 activity-score panel study and the external literature studies. The section split follows the scientific role of the data rather than the analyte split used for GOF diagnostics.

3.3.1 CYP2D6 DGI panel study

The CYP2D6 DGI panel study includes parent and metabolite plasma concentration-time profiles after oral clomiphene citrate administration in CYP2D6 activity-score groups. These profiles are used to assess whether the model captures activity-score-dependent differences in parent-metabolite disposition.

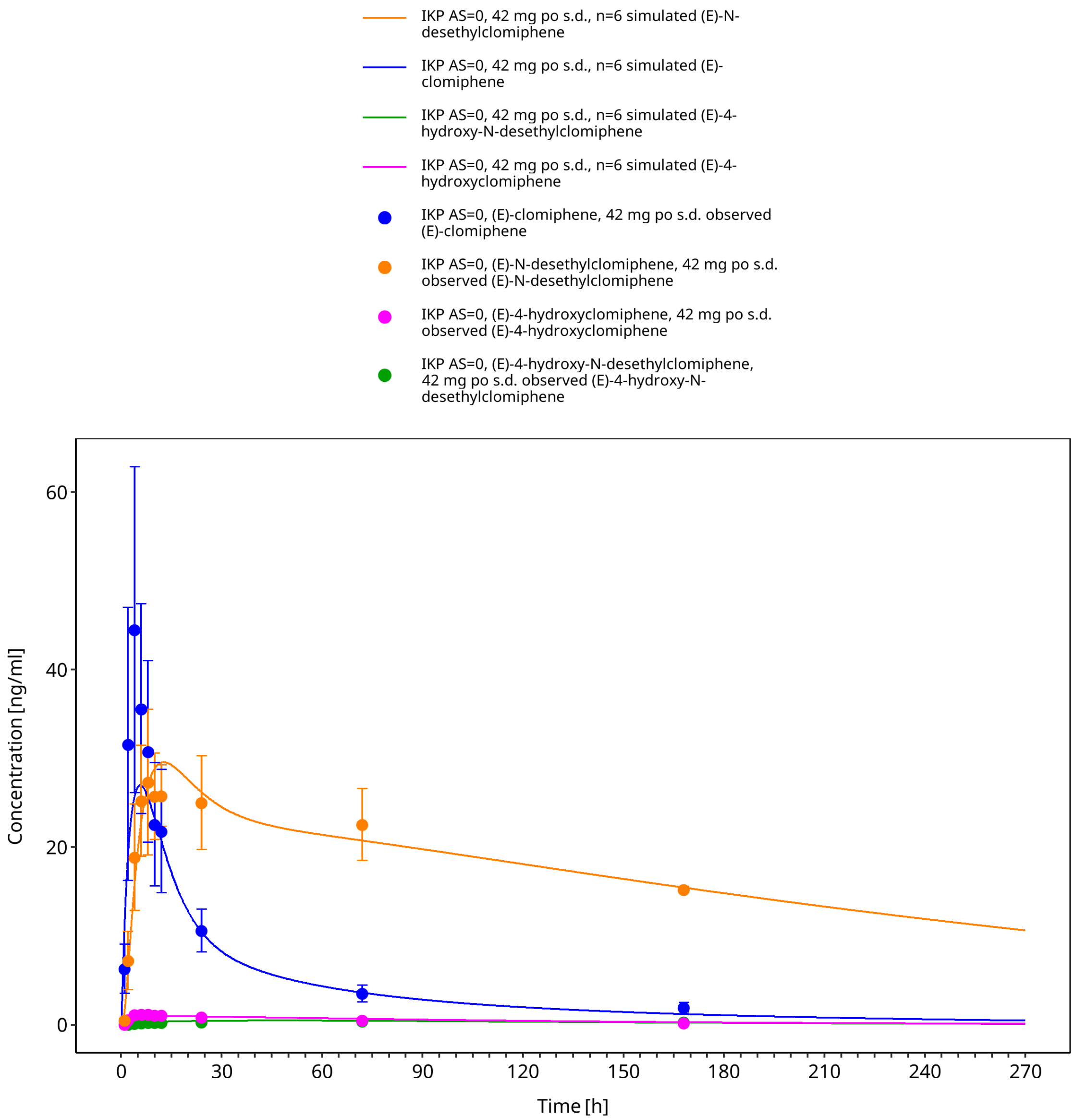

Figure 3-9: Time Profile Analysis

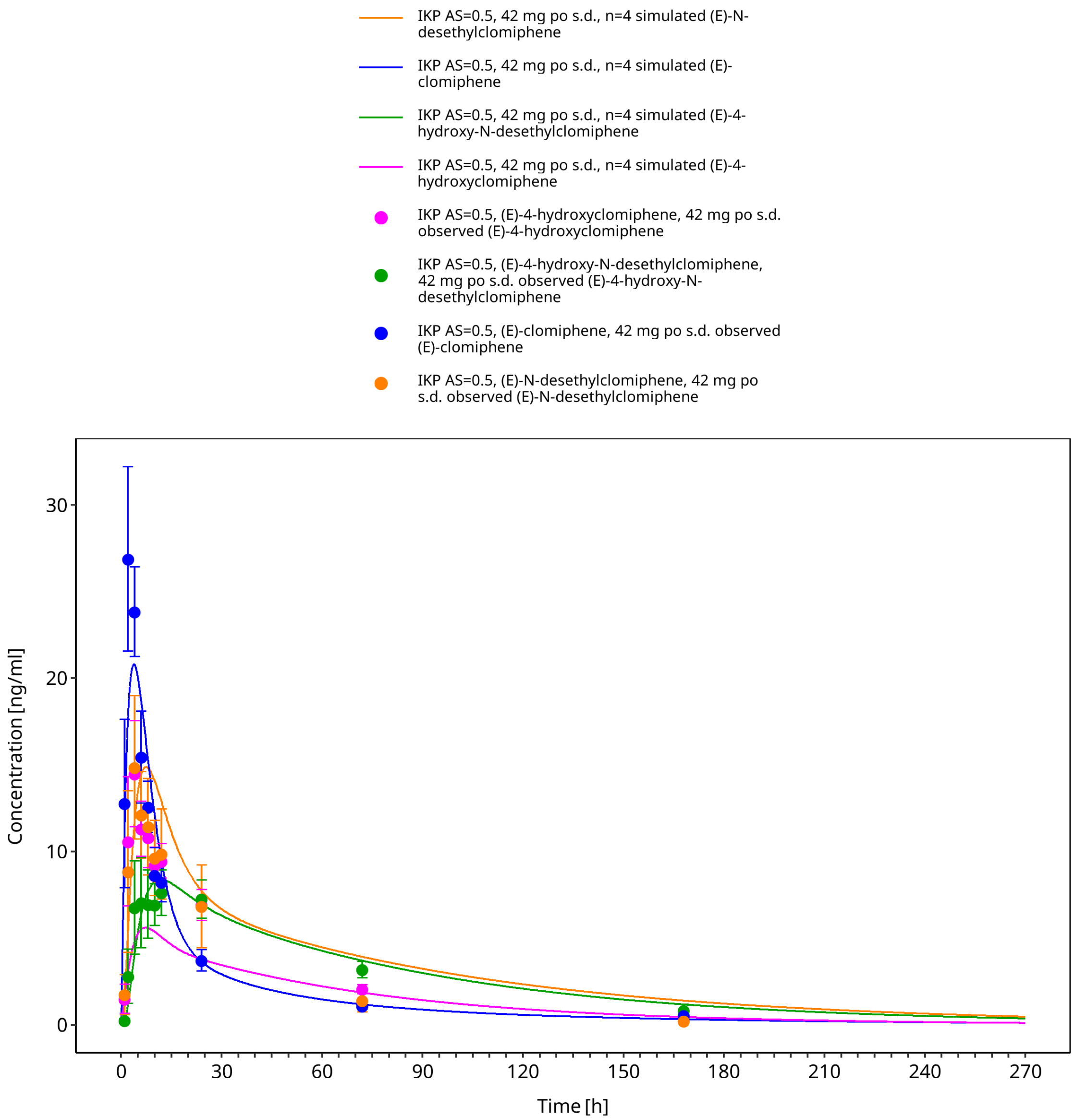

Figure 3-10: Time Profile Analysis

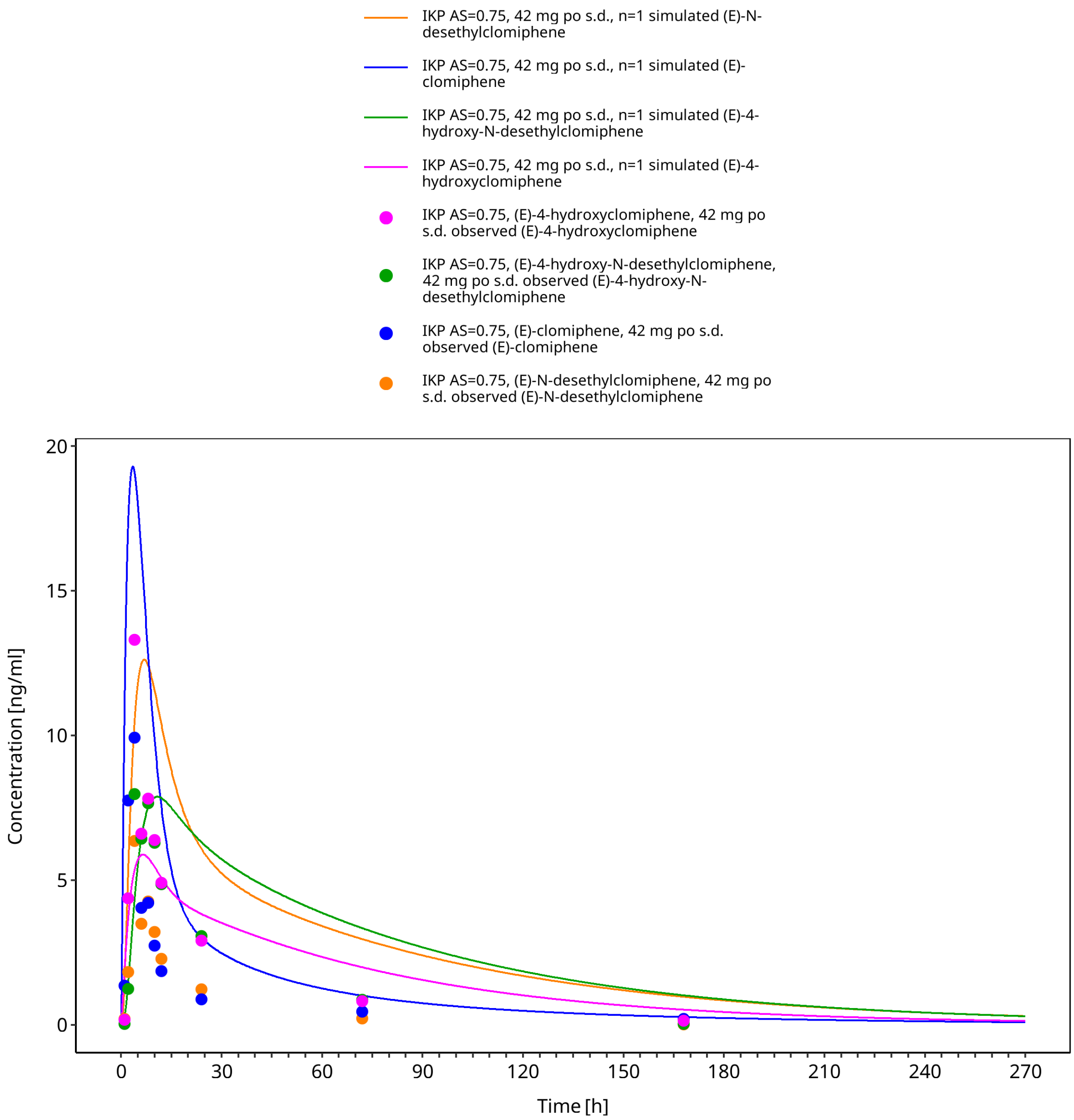

Figure 3-11: Time Profile Analysis

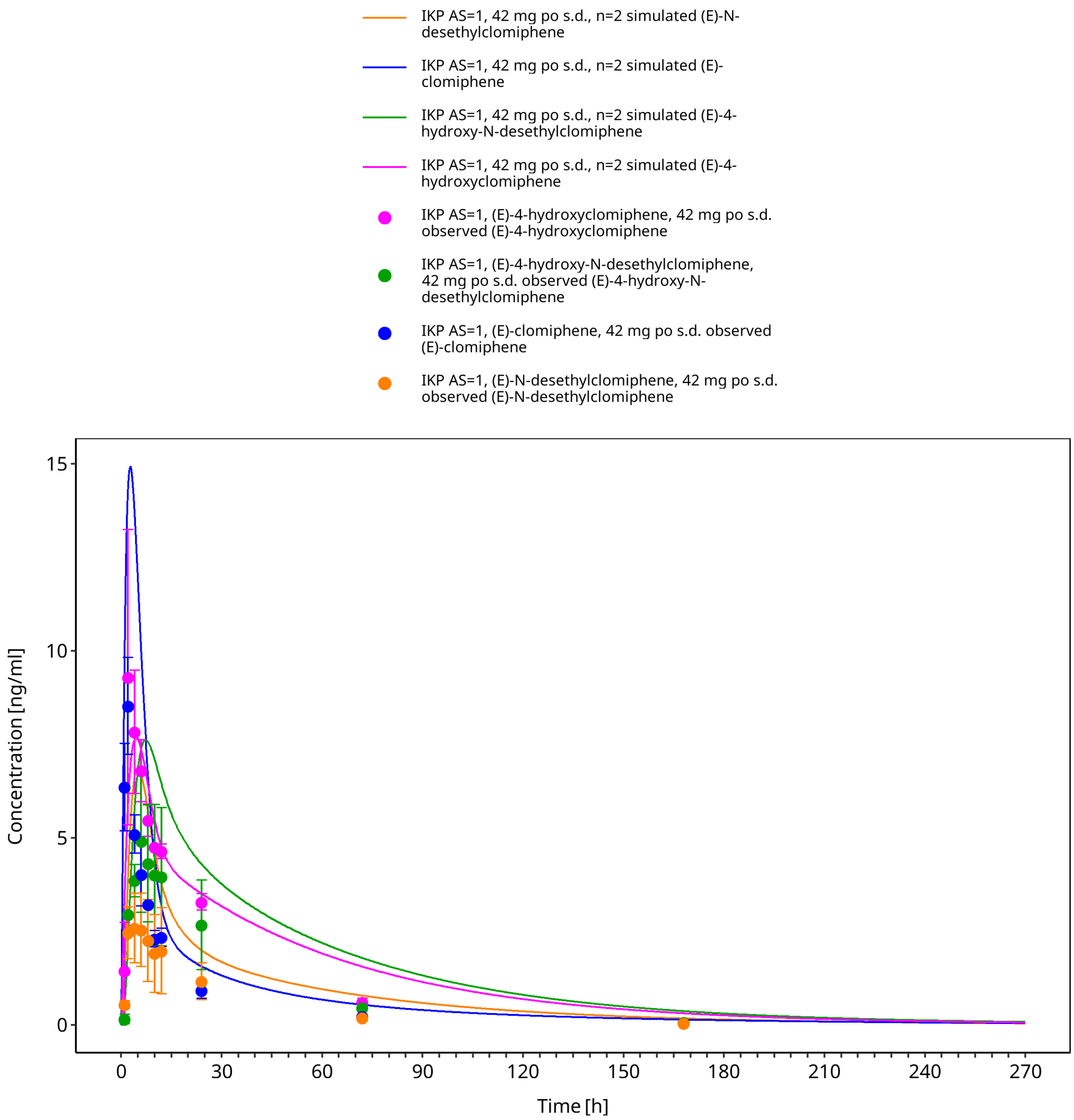

Figure 3-12: Time Profile Analysis

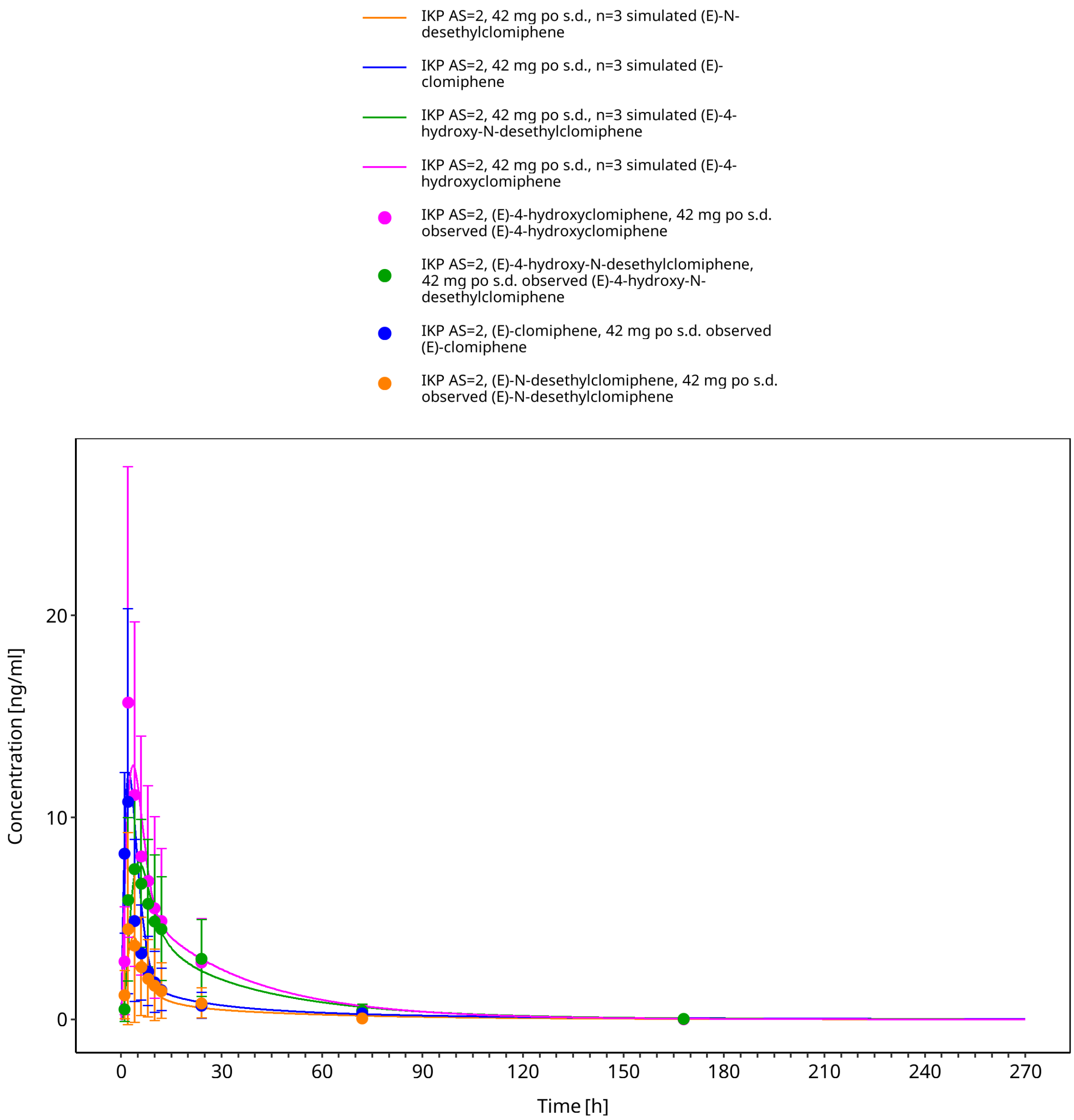

Figure 3-13: Time Profile Analysis

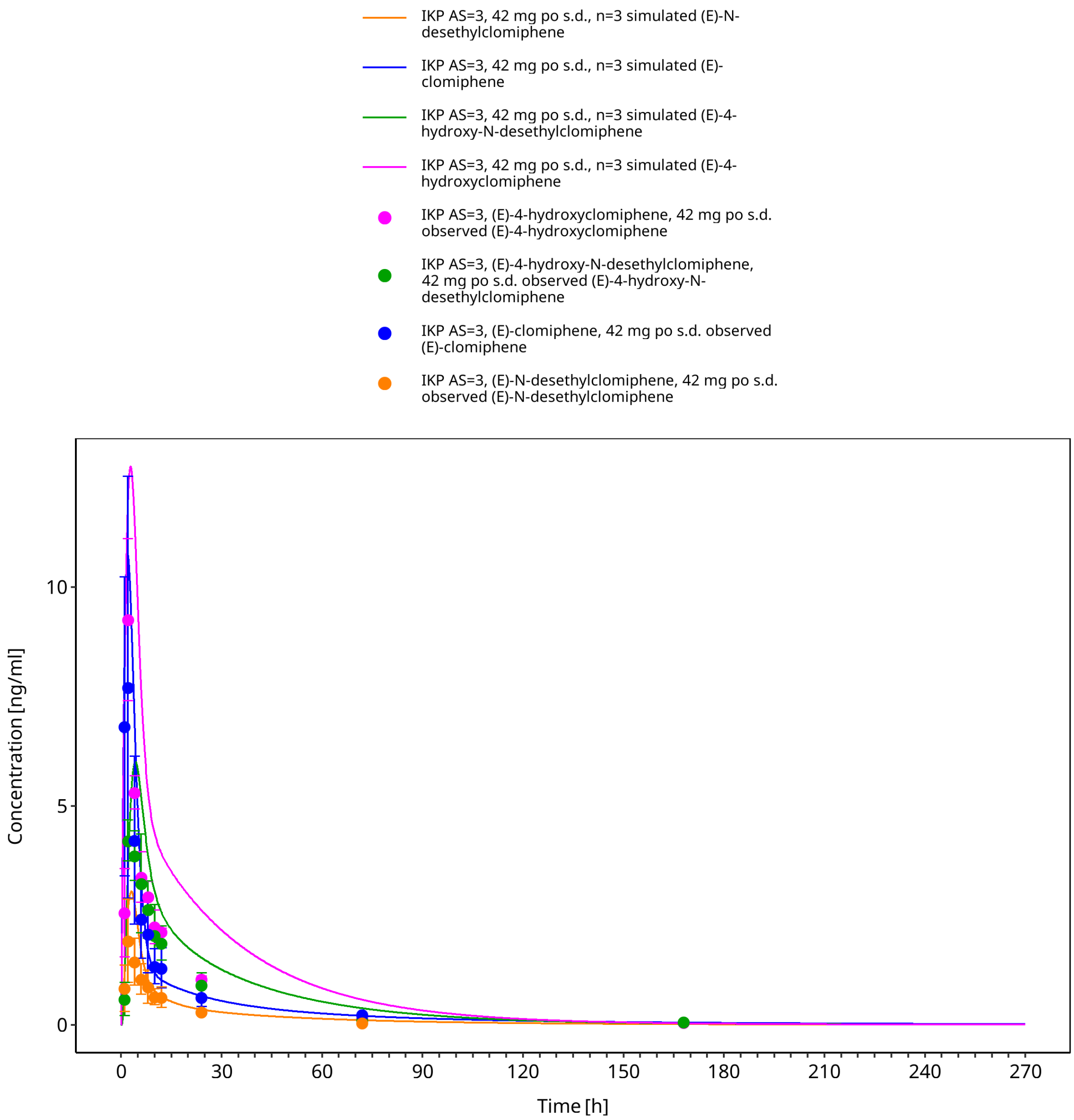

Figure 3-14: Time Profile Analysis

3.3.2 Literature studies

The literature study profiles include oral single-dose and multiple-dose (E)-clomiphene data. These profiles provide additional evaluation of parent-compound disposition across dosing conditions outside the CYP2D6 activity-score panel study.

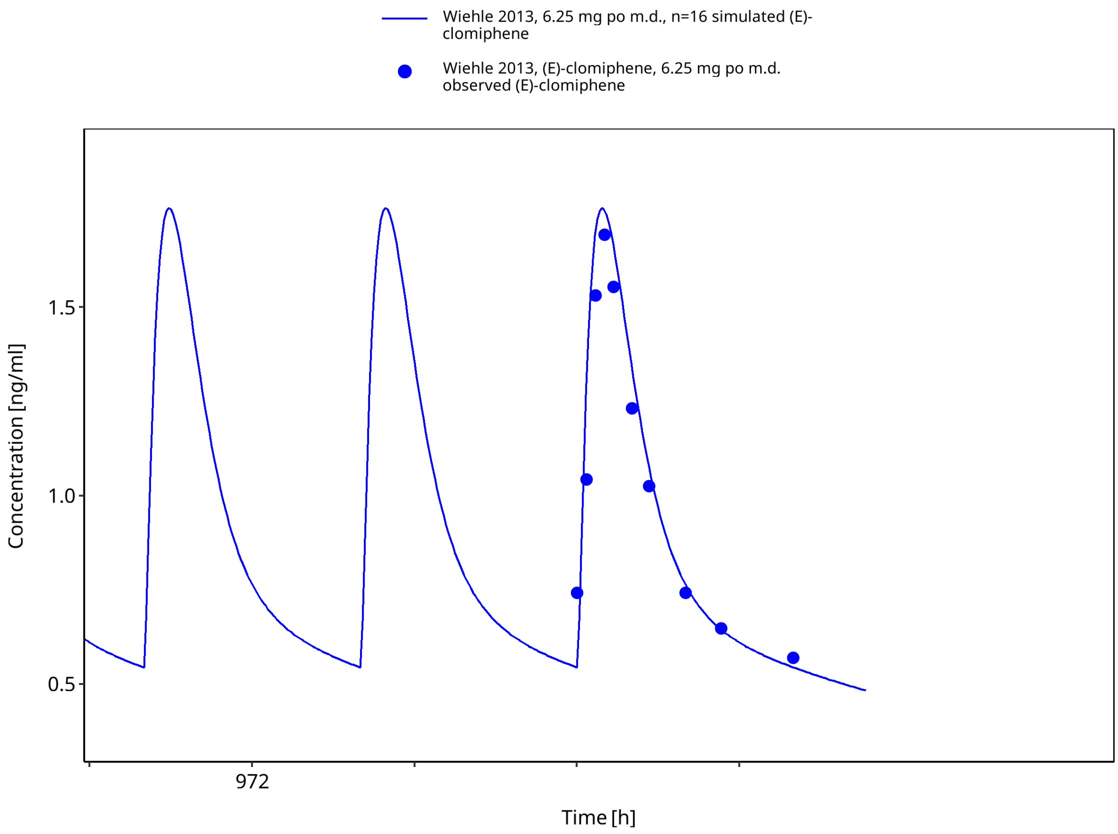

Figure 3-15: Time Profile Analysis

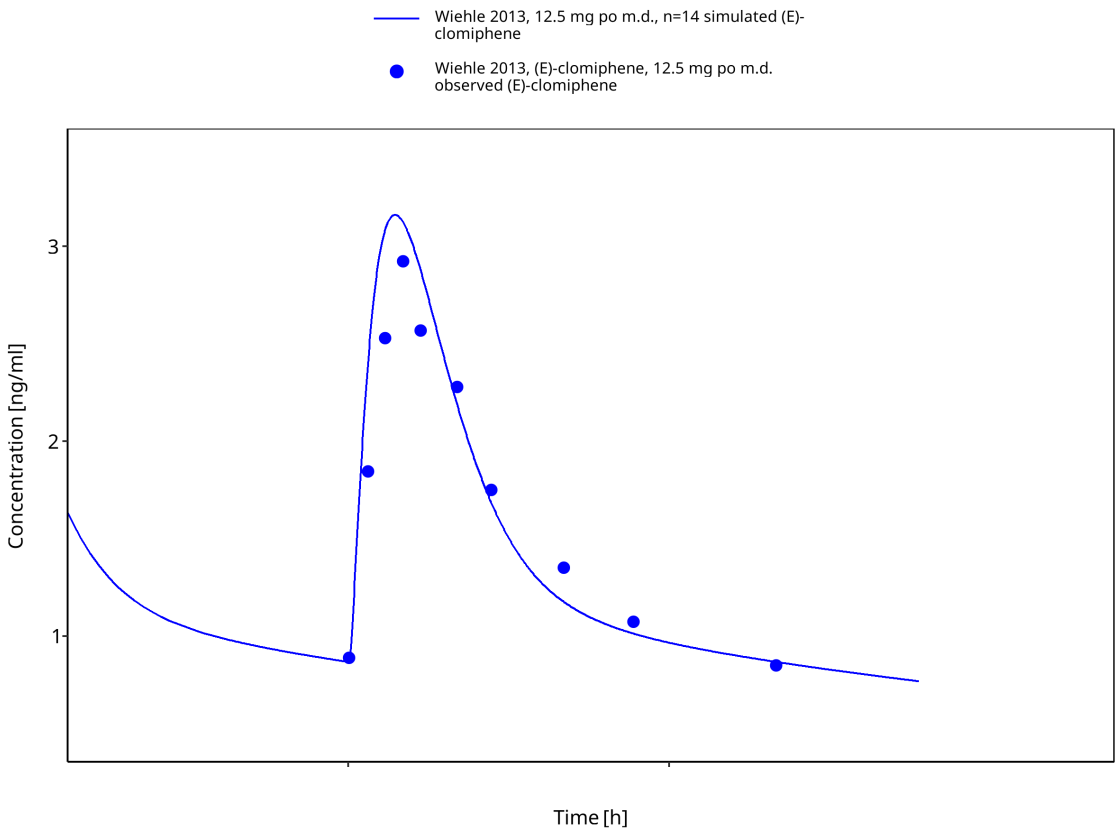

Figure 3-16: Time Profile Analysis

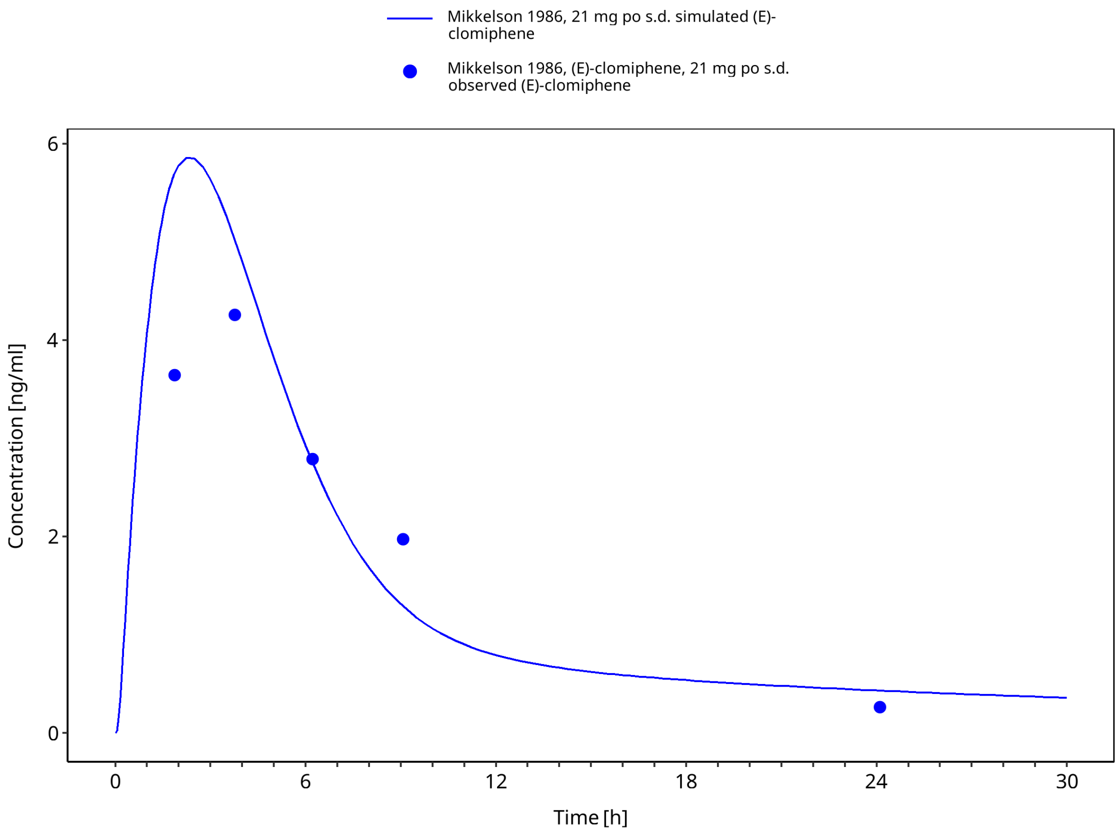

Figure 3-17: Time Profile Analysis

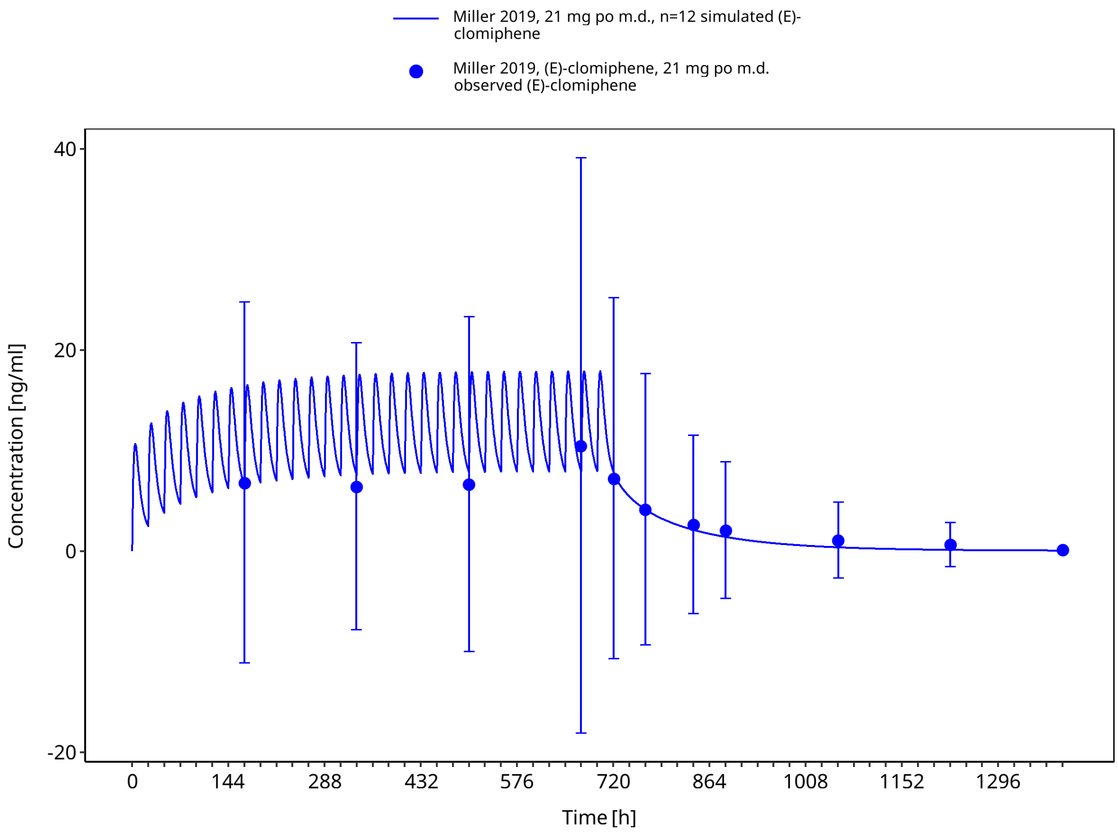

Figure 3-18: Time Profile Analysis

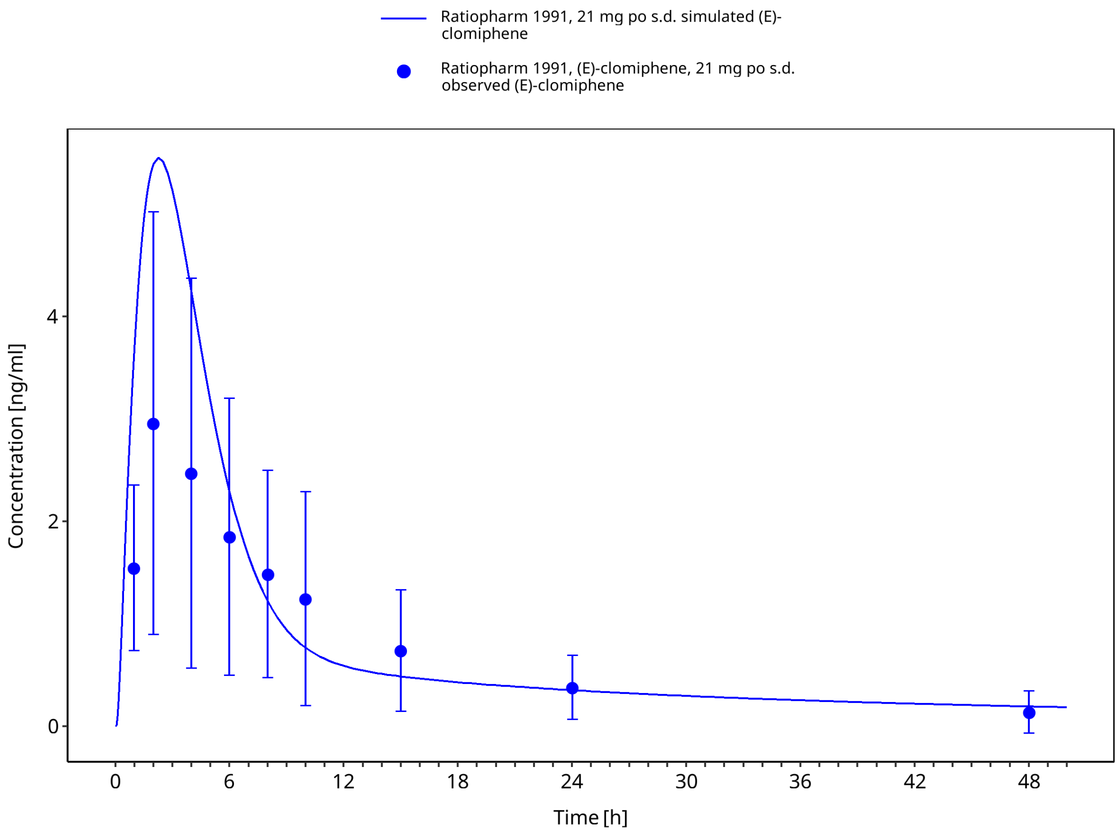

Figure 3-19: Time Profile Analysis

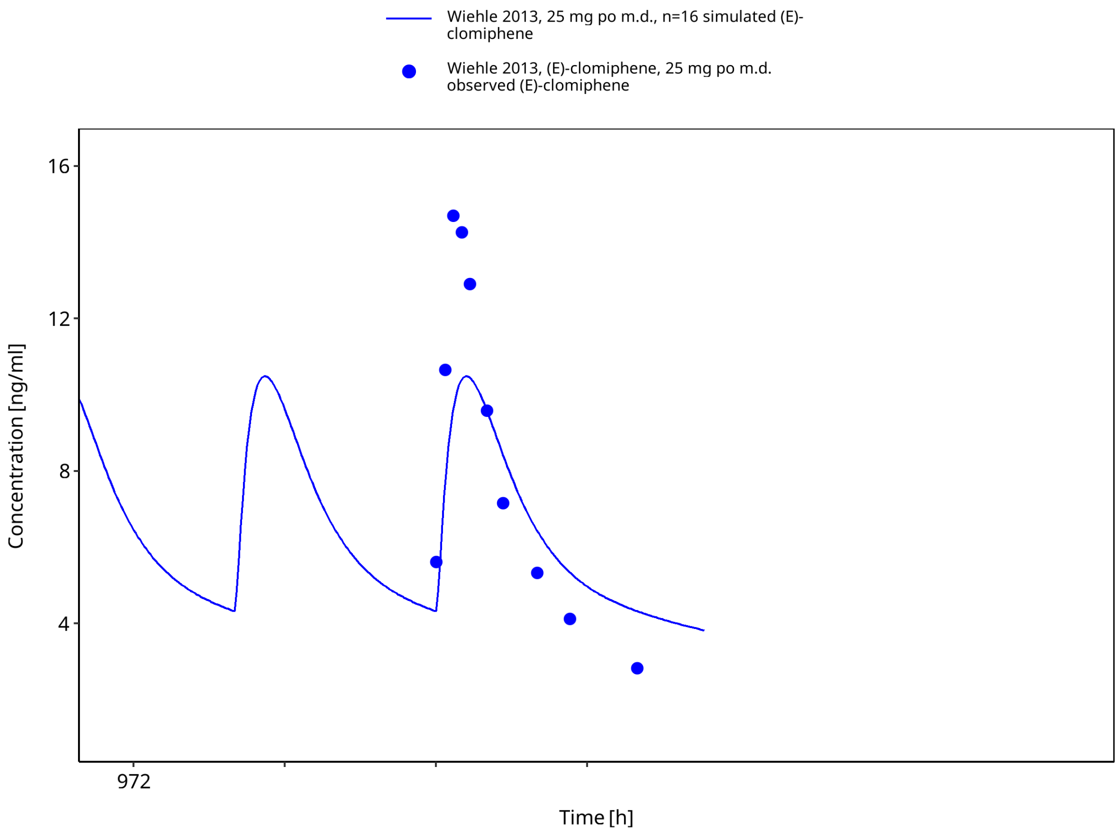

Figure 3-20: Time Profile Analysis

4 Conclusion

The clomiphene PBPK model describes (E)-clomiphene, (E)-N-desethylclomiphene, (E)-4-hydroxyclomiphene and (E)-4-hydroxy-N-desethylclomiphene plasma concentration-time profiles in adults after oral administration. The model includes activity-score-dependent CYP2D6 metabolism and supports evaluation of CYP2D6 drug-gene effects across the available adult clinical studies.

The report structure, diagnostic plots and repository layout are now aligned with the current OSP evaluation-report template. Final upload readiness still requires a user-side render and human QC of the rendered report, references and scientific conclusions.

5 References

[1] C Kovar, L Kovar, S Rüdesheim, D Selzer, B Ganchev, P Kröner, S Igel, R Kerb, E Schaeffeler, T E Mürdter, M Schwab, T Lehr. Prediction of Drug-Drug-Gene Interaction Scenarios of (E)-Clomiphene and Its Metabolites Using Physiologically Based Pharmacokinetic Modeling. Pharmaceutics, 2022;14:2604.

[2] S Rüdesheim, H L H Loer, D Feick, F Z Marok, L M Fuhr, D Selzer, D Teutonico, A R P Schneider, J Solodenko, S Frechen, M van der Lee, D J A R Moes, J J Swen, M Schwab, T Lehr. A Comprehensive CYP2D6 Drug-Drug-Gene Interaction Network for Application in Precision Dosing and Drug Development. Clin Pharmacol Ther, 2025.

[3] T J Mikkelson, P D Kroboth, W J Cameron, L W Dittert, V Chungi, P J Manberg. Single-dose pharmacokinetics of clomiphene citrate in normal volunteers. Fertil Steril, 1986;46:392-396.

[4] Ratiopharm GmbH. Clomifen-ratiopharm 50 mg Tabletten, Fachinformation, 2016.

[5] R Wiehle, G R Cunningham, N Pitteloud, J Wike, K Hsu, G K Fontenot, M Rosner, A Dwyer, J Podolski. Testosterone Restoration by Enclomiphene Citrate in Men with Secondary Hypogonadism: Pharmacodynamics and Pharmacokinetics. BJU Int, 2013;112:1188-1200.

[6] G D Miller, C Moore, V Nair, B Hill, S E Willick, A D Rogol, D Eichner. Hypothalamic-Pituitary-Testicular Axis Effects and Urinary Detection Following Clomiphene Administration in Males. J Clin Endocrinol Metab, 2019;104:906-914.

[7] T E Mürdter, R Kerb, M Turpeinen, W Schroth, B Ganchev, G M Böhmer, S Igel, E Schaeffeler, U Zanger, H Brauch, M Schwab. Genetic polymorphism of cytochrome P450 2D6 determines oestrogen receptor activity of the major infertility drug clomiphene via its active metabolites. Hum Mol Genet, 2012;21:1145-1154.

[8] P Kröner, J Zolk, T E Mürdter, M Schwab, T Lehr. Determination of clomiphene and active metabolites in human plasma by LC-MS/MS. Talanta, 2021;221:121658.

[9] M Mazzarino, M Biava, X de la Torre, I Fiacco, F Botrè. Characterization of biotransformation pathways of clomiphene, tamoxifen and toremifene as assessed by LC-MS/(MS) following in vitro and excretion studies. Anal Bioanal Chem, 2013;405:5467-5487.

[10] R Watanabe, T Esaki, H Kawashima, Y Natsume-Kitatani, C Nagao, R Ohashi, K Mizuguchi. Predicting Fraction Unbound in Human Plasma from Chemical Structure: Improved Accuracy in the Low Value Ranges. Mol Pharm, 2018;15:5302-5311.

[11] W Schmitt. General approach for the calculation of tissue to plasma partition coefficients. Toxicol In Vitro, 2008;22:457-467.

[12] T Rodgers, D Leahy, M Rowland. Physiologically based pharmacokinetic modeling 1: predicting the tissue distribution of moderate-to-strong bases. J Pharm Sci, 2005;94:1259-1276.

[13] T Rodgers, M Rowland. Physiologically based pharmacokinetic modelling 2: predicting the tissue distribution of acids, very weak bases, neutrals and zwitterions. J Pharm Sci, 2006;95:1238-1257.

[14] L M Berezhkovskiy. Volume of distribution at steady state for a linear pharmacokinetic system with peripheral elimination. J Pharm Sci, 2004;93:1628-1640.